

# ROTAX 912 iS ILLUSTRATED PARTS CATALOG & CAD COMPONENT MAPPING

**\*\*Document Ref: IPC-912iS-ED03\*\***  
**\*\*CAD Hierarchy, 3D Mesh Identifiers & Torque Specifications\*\***  
---

## 1. 3D CAD MESH & FAULT CORRELATION TABLE

| Component Name          | 3D Mesh Identifier (Blender / Unity) | ATA Chapter | Subsystem Tag                 |
|-------------------------|--------------------------------------|-------------|-------------------------------|
| Engine Shroud & Baffles | `Covers_Theme_M_PlasticTheme_0`      | ATA 72-00   | `ENGINE_CORE_COOLING`         |
| Fuel Rail & Injectors   | `Rotax_912i_Base_M_PlasticGreen_0`   | ATA 73-10   | `FUEL_INJECTION_SYSTEM`       |
| Wiring Harness & Coils  | `Wiring_Harness_M_Copper_0`          | ATA 74-20   | `IGNITION_SYSTEM`             |
| Oil Reservoir & Tank    | `Oil_Tank_M_Steel_0`                 | ATA 79-00   | `LUBRICATION_SYSTEM`          |
| Reduction Gearbox       | `Gearbox_Type_2_M_Steel_0`           | ATA 72-10   | `REDUCTION_GEARBOX`           |
| Exhaust Manifold        | `Exhaust_System_M_SteelDark_0`       | ATA 78-10   | `EXHAUST_SYSTEM`              |
| External Alternator     | `External_Alternator_M_GreyDark_0`   | ATA 24-00   | `ELECTRICAL_POWER_GENERATION` |
| Dual FADEC ECU          | `ECU_M_PlasticDark_0`                | ATA 76-00   | `ENGINE_CONTROL_UNIT`         |

---

## 2. STANDARD FASTENER TORQUE SPECIFICATIONS

- \* **Cylinder Head Stud Nuts (M8):** 22 Nm (195 in. lb) followed by 180 deg turn.
- \* **Spark Plugs (M12 x 1.25):** 20 Nm (177 in. lb) cold engine.
- \* **Exhaust Flange Nuts (M8):** 15 Nm (133 in. lb) with high-temp anti-seize paste.
- \* **Oil Drain Magnetic Plug (M12 x 1.5):** 25 Nm (221 in. lb).
- \* **Fuel Rail Retaining Screws (M5):** 6 Nm (53 in. lb).
- \* **Gearbox Housing Bolts (M6):** 10 Nm (89 in. lb) with Loctite 243.
- \* **Alternator Mounting Bracket (M8):** 25 Nm (221 in. lb).